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DC PANEL METER Using Arduino

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Panel meters in regulated power supplies are used to display electrical parameters like voltage and current. Presented here is a circuit to display DC voltage and current of power supplies, including DIY-type ones.

Circuit and working

Circuit diagram of the DC panel meter using Arduino is shown in Fig. 1. The circuit is built around Arduino Uno board (Board1), current sensor IC ACS712 (IC1), 16 × 2 LCD (LCD1), npn transistor BC547 (T1), piezo buzzer (PZ1) and a few other components.

Power supply voltage is sampled using the voltage divider network built around resistors R1 and R2. Current consumed by the load is sampled using ACS712. Sampled voltage and current are given to Arduino, which is the brain of the circuit and performs calculations for displaying the parameters on the LCD1.

Voltage at Arduino analogue and digital pins should not exceed 5V. Hence, an alert message is displayed on the LCD1 and the buzzer sounds whenever sampled voltage and current go beyond the limit.

Arduino Uno. Arduino Uno is a popular open source microcontroller (MCU) development board based on ATmega328P MCU. It has 14 digital input/output (I/O) pins (of which six can be used as PWM outputs), six analogue inputs, 16MHz crystal, USB connector, power jack and ICSP header. It comes preloaded with Arduino bootloader.

ACS712. Current sensor IC ACS712 accurately detects AC or DC current, and produces analogue voltage proportional to the current passing through it. Its internal views are shown in Fig. 2. The device consists of a precise, low-offset, linear Hall circuit with a copper conduction path located near the surface of the die. Applied current flowing through the copper conduction path generates a magnetic field, which the Hall IC converts into proportional voltage. Device accuracy is optimised through the close proximity of the magnetic signal to the Hall transducer.

Output of the IC is 2.5V when no current is passed through it—it swings around 2.5V at the rate of 185mV/A for 5A version, 100mV/A for 20A version and 66mV/A for 30A version. The graph shown in Fig. 3 shows output voltage versus sensed current of 5A version sensor IC. A readily-available 5A version sensor module is used in this circuit.

Internal resistance of the conductive path is 1.2mΩ, providing low power loss. Terminals of the conductive path (pins 1 to 4) are electrically-isolated from signal leads (pins 5 through 8). This allows ACS712 to be used in applications that require electrical isolation without the use of opto-isolators or any other costly isolation technique.

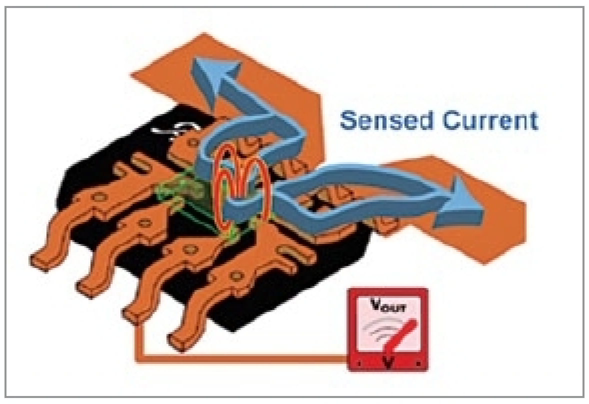

Fig. 2: Internal view of ACS712

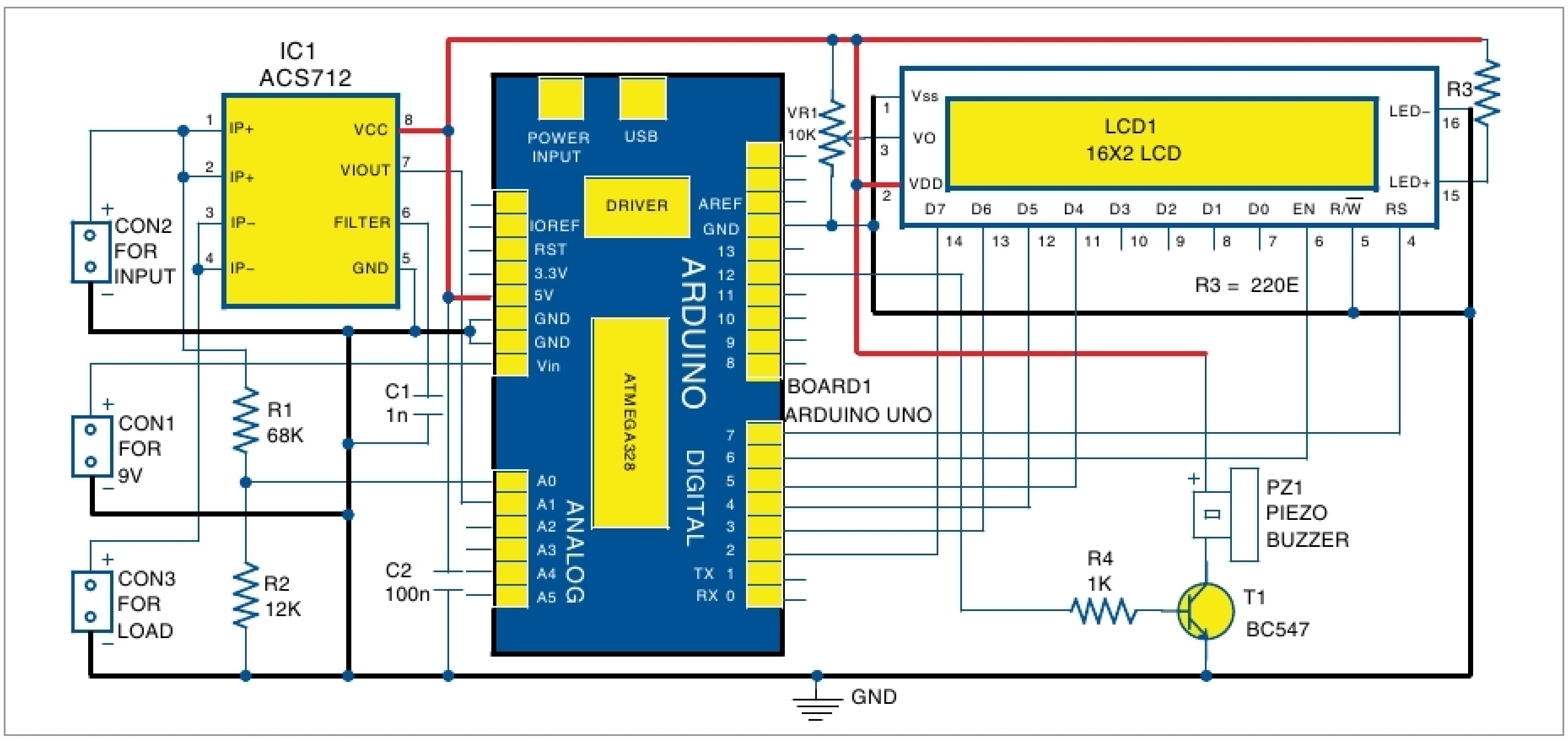

Fig. 1: Circuit diagram of DC panel meter using Arduino